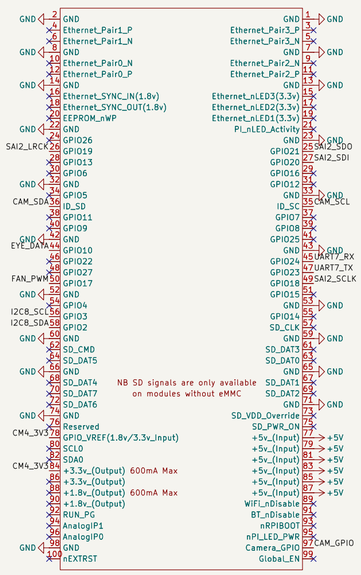
Radxa CM4 on DF40C sockets

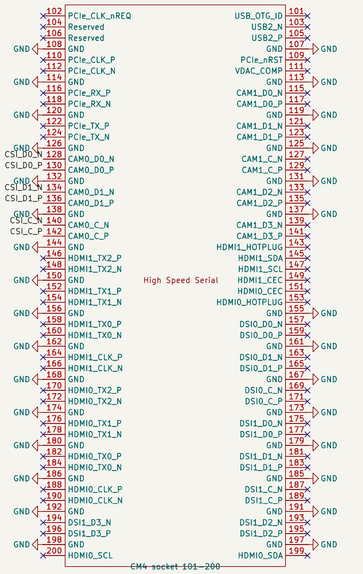
Pin allocation: docs/head-pcb.md section 4. Unused pins not connected, as on the NANO-A

J9



CM4 socket 1–100

J10



CM4 socket 101–200

GPIO_VREF (78) on the module's own 3.3 V output (84; 86 is the same rail, left open): 3.3 V GPIO, as the NANO-A's H2 jumper

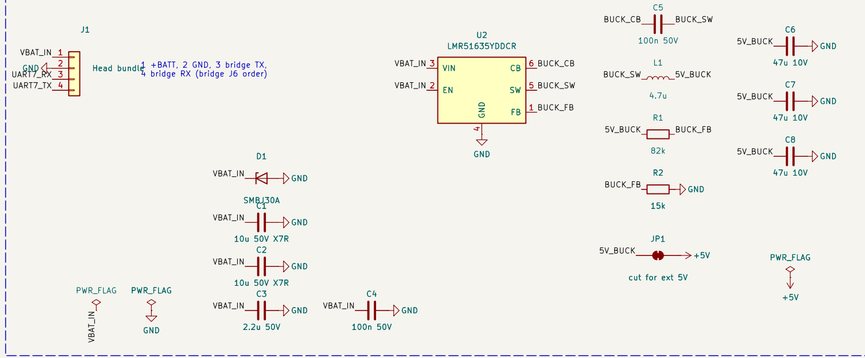
H5



Radxa CM4

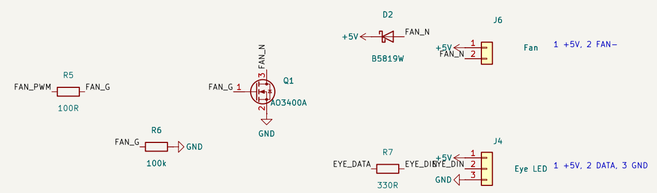
Head bundle in, 5 V buck

LMR51635Y, 1.1 MHz, 4.7 uH (Ti table 8-2: 3.3 uH), 82k/15k sets 5.17 V, EN tied to VIN. F6 (2 A) on the bridge fuses this feed. JP1 joins the buck to the 5 V rail: cut it to feed 5 V from outside (J12). The feedback divider stays on the buck side



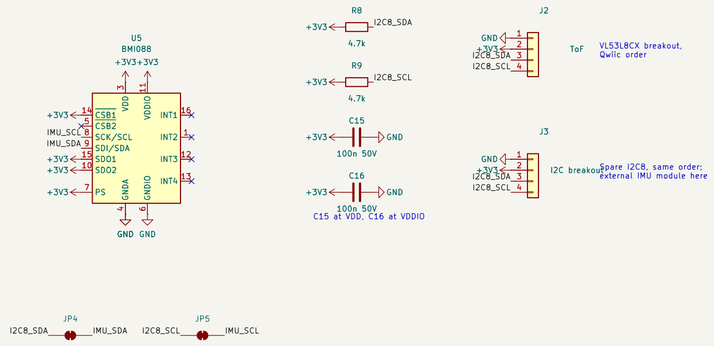
Fan and eye LED

Fan: low side, PWM1_CH2 (pin 50). Eye: WS2812-type, SPI1 MOSI (pin 44), 3.3 V data



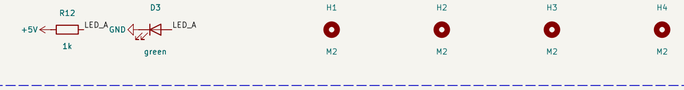
I2C8: head IMU, ToF, breakout

BMI088 on the board. I2C: PS and CSB1 high, CSB2 open (allowed in I2C mode): SDO1 and SDO2 high: accelerometer 0x19, gyroscope 0x69 (strapped so the pin rows need no vias between alternate pins). R8, R9: the bus pull-ups (the ToF breakout adds 10k). Cut JP4 and JP5 to use an external IMU module on J3



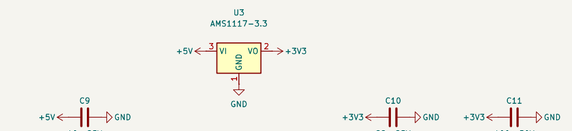
Power LED, mounting holes

M2 holes, one per corner, not connected



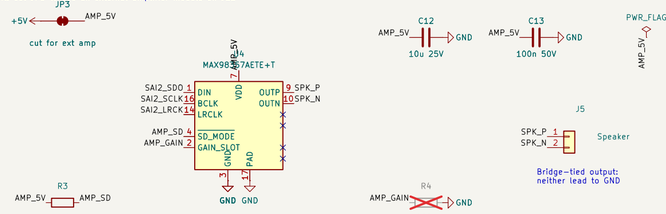
3.3 V

AMS1117-3.3 from 5 V, as on the NANO-A. Camera, ToF, IMU, mic, I2C breakout



Speaker amplifier

MAX98357A on SAI2. SD_MODE 1M to 5 V: (L+R)/2 mono. GAIN open = 9 dB: fit R4 (100k to GND) for 15 dB. JP3 feeds it: cut JP3 to use an external amplifier module on J11

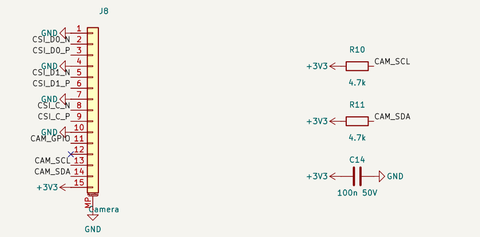


Microphone

INMP441 on SAI2. L/R tied to GND on the mic side (left slot)

Camera FPC

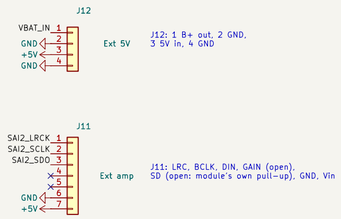
Pi 15-pin camera order. CAM0 lanes (Radxa CSI1), I2C6 with 4.7k pull-ups, CAM_GPIO enable. 100 ohm pairs



Pin 12 (camera clock) not connected: the module has its own oscillator

External fallbacks

Wire pads, 2.54 mm. J12: cut JP1 and feed 5 V from an external step-down. J11: cut JP3 and wire an amplifier module, pins in the generic module's order. J3 (section 6) takes an external IMU module



Bootstrapped by scripts/kicad/gen_head_sch.py; edit this file from now on
Design note: docs/head-pcb.md

Sheet: /
File: head.kicad_sch

Title: Dukki head carrier

Size: A2 | Date: 12/10/2026 | **Rev: A, draft**
KiCad E.D.A. 10.0.6 | Id: 1/1